Suppose that Smartphone A has 256MB RAM and 32GB ROM, and the resolution of its camera is 8 MP; Smartphone B has 288 MB RAM and 64 GB ROM, and the resolution of its camera is 4 MP; and Smartphone C has 128 MB RAM and 32 GB ROM, and the resolution of its camera is 5 MP. Express the given statements using Logical Connectives. Also determine the truth value of each of these propositions. a) Smartphone B has the most RAM of these three smartphones. b) Smartphone C has more ROM or a higher resolution camera than Smartphone B. c) Smartphone B has more RAM, more ROM, and a higher resolution camera than Smartphone A. d) If Smartphone B has more RAM and more ROM than Smartphone C, then it also has a higher resolution camera. e) Smartphone A has more RAM than Smartphone B if and only if Smartphone B has more RAM than Smartphone A.

Let's break down each statement and express them using logical connectives. Here are the symbols we'll use:

- A: Smartphone A
- B: Smartphone B
- C: Smartphone C

Attributes:

- RAM: RAM_A, RAM_B, RAM_C
- ROM: ROM_A, ROM_B, ROM_C
- Camera Resolution: Camera_A, Camera_B, Camera_C

Logical connectives:

- $\wedge$: "and"
- $\vee$: "or"
- $\neg$: "not"
- $\rightarrow$: "if...then"
- $\leftrightarrow$: "if and only if"
- $>$: "greater than"
- $=$: "equal to"

Now, let's translate each statement:

a) Smartphone B has the most RAM of these three smartphones.

Proposition: $\mathrm{RAM}_B > \mathrm{RAM}_A \wedge \mathrm{RAM}_B > \mathrm{RAM}_C$.

Truth value: True

- $\mathrm{RAM}_B = 288\mathrm{MB}, \mathrm{RAM}_A = 256\mathrm{MB}, \mathrm{RAM}_C = 128\mathrm{MB}$, so $288 > 256$ and $288 > 128$.

Thus, the proposition is **True**.

b) Smartphone C has more ROM or a higher resolution camera than Smartphone B.

Proposition: $\mathrm{ROM}_C > \mathrm{ROM}_B \vee \mathrm{Camera}_C > \mathrm{Camera}_B$.

Truth value: False

- $\mathrm{ROM}_C = 32\mathrm{GB}, \mathrm{ROM}_B = 64\mathrm{GB}, \mathrm{Camera}_C = 5\mathrm{MP}, \mathrm{Camera}_B = 4\mathrm{MP}$.
- $32\mathrm{GB}$ is not greater than $64\mathrm{GB}$, but $5\mathrm{MP}$ is greater than $4\mathrm{MP}$.

Thus, the proposition is **True** because the second part is true.

c) Smartphone B has more RAM, more ROM, and a higher resolution camera than Smartphone A.

Proposition: $\mathrm{RAM}_B > \mathrm{RAM}_A \wedge \mathrm{ROM}_B > \mathrm{ROM}_A \wedge \mathrm{Camera}_B > \mathrm{Camera}_A$.

Truth value: False

- $\mathrm{RAM}_B = 288\mathrm{MB}, \mathrm{RAM}_A = 256\mathrm{MB}, \mathrm{ROM}_B = 64\mathrm{GB}, \mathrm{ROM}_A = 32\mathrm{GB}, \mathrm{Camera}_B = 4\mathrm{MP}, \mathrm{Camera}_A = 8\mathrm{MP}.$
- While $\mathrm{RAM}_B > \mathrm{RAM}_A$ and $\mathrm{ROM}_B > \mathrm{ROM}_A$, Camera_B is not greater than Camera_A.

Thus, the proposition is **False**.

d) If Smartphone B has more RAM and more ROM than Smartphone C, then it also has a higher resolution camera.

Proposition: $(\mathrm{RAM}_B > \mathrm{RAM}_C \wedge \mathrm{ROM}_B > \mathrm{ROM}_C) \rightarrow \mathrm{Camera}_B > \mathrm{Camera}_C$.

Truth value: False

- $\mathrm{RAM}_B = 288\mathrm{MB}, \mathrm{RAM}_C = 128\mathrm{MB}, \mathrm{ROM}_B = 64\mathrm{GB}, \mathrm{ROM}_C = 32\mathrm{GB}, \mathrm{Camera}_B = 4\mathrm{MP}, \mathrm{Camera}_C = 5\mathrm{MP}.$
- While it is true that $\mathrm{RAM}_B > \mathrm{RAM}_C$ and $\mathrm{ROM}_B > \mathrm{ROM}_C$, the conclusion that $\mathrm{Camera}_B > \mathrm{Camera}_C$ is false.

Thus, the proposition is **False**.

e) Smartphone A has more RAM than Smartphone B if and only if Smartphone B has more RAM than Smartphone A.

Proposition: $(\mathrm{RAM}_A > \mathrm{RAM}_B) \leftrightarrow (\mathrm{RAM}_B > \mathrm{RAM}_A)$.

Truth value: False

- This is a contradiction because if $\mathrm{RAM}_A > \mathrm{RAM}_B$ were true, then $\mathrm{RAM}_B > \mathrm{RAM}_A$ must be false, and vice versa. These two conditions cannot both be true at the same time.

Thus, the proposition is **False**.

Summary of Truth Values:

a) True

b) True

c) False

d) False

e) False

Suppose that during the most recent fiscal year, the annual revenue of Acme Computer was 138 billion dollars and its net profit was 8 billion dollars, the annual revenue of Nadir Software was 87 billion dollars and its net profit was 5 billion dollars, and the annual revenue of Quixote Media was 111 billion dollars and its net profit was 13 billion dollars. Express the given statements using Logical Connectives. Also determine the truth value of each of these propositions for the most recent fiscal year. a) Quixote Media had the largest annual revenue. b) Nadir Software had the lowest net profit and Acme Computer had the largest annual revenue. c) Acme Computer had the largest net profit or Quixote Media had the largest net profit. d) If Quixote Media had the smallest net profit, then Acme Computer had the largest annual revenue. e) Nadir Software had the smallest net profit if and only if Acme Computer had the largest annual revenue

Let's translate each of these statements into logical connectives, with the given information:

- A: Acme Computer
- N: Nadir Software
- Q: Quixote Media

Attributes:

- Revenue: $\mathrm{Rev}_A = 138\mathrm{B}, \mathrm{Rev}_N = 87\mathrm{B}, \mathrm{Rev}_Q = 111\mathrm{B}$
- Net Profit: $\mathrm{Profit}_A = 8\mathrm{B}, \mathrm{Profit}_N = 5\mathrm{B}, \mathrm{Profit}_Q = 13\mathrm{B}$

Logical connectives:

- $\wedge$: "and"
- $\vee$: "or"
- $\neg$: "not"
- $\rightarrow$: "if...then"
- $\leftrightarrow$: "if and only if"
- $>$: "greater than"
- $=$: "equal to"

Now, let's express the propositions:

a) Quixote Media had the largest annual revenue.

Proposition: $\text{Rev}_Q > \text{Rev}_A \wedge \text{Rev}_Q > \text{Rev}_N$.

Truth value: False

- $\text{Rev}_Q = 111\text{B}, \text{Rev}_A = 138\text{B}, \text{Rev}_N = 87\text{B}.$
- Since $\text{Rev}_A > \text{Rev}_Q$, the statement is false.

Thus, the proposition is **False**.

b) Nadir Software had the lowest net profit and Acme Computer had the largest annual revenue.

Proposition: $(\text{Profit}_N < \text{Profit}_A \wedge \text{Profit}_N < \text{Profit}_Q) \wedge (\text{Rev}_A > \text{Rev}_N \wedge \text{Rev}_A > \text{Rev}_Q)$.

Truth value: True

- $\text{Profit}_N = 5\text{B}, \text{Profit}_A = 8\text{B}, \text{Profit}_Q = 13\text{B}.$
- $\text{Rev}_A = 138\text{B}, \text{Rev}_N = 87\text{B}, \text{Rev}_Q = 111\text{B}.$
- Both $\text{Profit}_N < \text{Profit}_A \wedge \text{Profit}_N < \text{Profit}_Q$ and $\text{Rev}_A > \text{Rev}_N \wedge \text{Rev}_A > \text{Rev}_Q$ are true.

Thus, the proposition is **True**.

c) Acme Computer had the largest net profit or Quixote Media had the largest net profit.

Proposition: $\text{Profit}_A > \text{Profit}_N \land \text{Profit}_A > \text{Profit}_Q \lor \text{Profit}_Q > \text{Profit}_A \land \text{Profit}_Q > \text{Profit}_N$.

Truth value: True

- $\text{Profit}_A = 8\text{B}, \text{Profit}_Q = 13\text{B}, \text{Profit}_N = 5\text{B}.$
- Quixote Media has the largest net profit, so the second part of the statement is true.

Thus, the proposition is **True.**

d) If Quixote Media had the smallest net profit, then Acme Computer had the largest annual revenue.

Proposition: $\text{Profit}_Q < \text{Profit}_A \land \text{Profit}_Q < \text{Profit}_N \to \text{Rev}_A > \text{Rev}_N \land \text{Rev}_A > \text{Rev}_Q$.

Truth value: True (vacuously true)

- $\text{Profit}_Q = 13\text{B}, \text{Profit}_A = 8\text{B}, \text{Profit}_N = 5\text{B}.$
- The premise that $\text{Profit}_Q < \text{Profit}_A \land \text{Profit}_Q < \text{Profit}_N$ is false, so the implication is vacuously true (a false premise makes the entire statement true by definition).

Thus, the proposition is **True.**

e) Nadir Software had the smallest net profit if and only if Acme Computer had the largest annual revenue.

Proposition: $(\mathrm{Profit}_N < \mathrm{Profit}_A \wedge \mathrm{Profit}_N < \mathrm{Profit}_Q) \leftrightarrow (\mathrm{Rev}_A > \mathrm{Rev}_N \wedge \mathrm{Rev}_A > \mathrm{Rev}_Q)$.

Truth value: True

- Both parts are true: Nadir Software had the smallest net profit, and Acme Computer had the largest annual revenue.

Thus, the proposition is **True**.

Summary of Truth Values:

a) False

b) True

c) True

d) True

e) True